

Neural Networks and Deep Learning Lab-3

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1 Introduction

Loss functions play a crucial role in training regression models, as they define how prediction errors are quantified and minimized. Different loss functions exhibit different optimization behaviors, affecting convergence speed, numerical stability, and robustness to outliers.

In this assignment, I studied and compared three commonly used loss functions:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Huber Loss

The goal is to experimentally analyze:

- Which loss function converges faster and why?
- Which loss function is most robust to outliers?

2 Dataset and Preprocessing

I used the California Housing dataset, a real-world regression dataset containing eight numerical features. The target variable represents the median house value.

3 Loss Functions

3.1 Mean Squared Error (MSE)

$$L_{\text{MSE}} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE penalizes large errors heavily due to squaring, leading to fast convergence but high sensitivity to outliers.

3.2 Mean Absolute Error (MAE)

$$L_{\text{MAE}} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE grows linearly with error magnitude, making it robust to outliers. However, its non-smooth gradient leads to slower convergence.

3.3 Huber Loss

$$L_{\text{Huber}} = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

Huber loss combines the advantages of MSE and MAE by being quadratic for small errors and linear for large errors.

4 Training Methodology

All models are trained using batch Gradient Descent. A gradient-norm based stopping criterion is used:

$$\|\nabla L\|_2 < \epsilon$$

This ensures fair convergence comparison across loss functions with different scales.

5 Results and Analysis

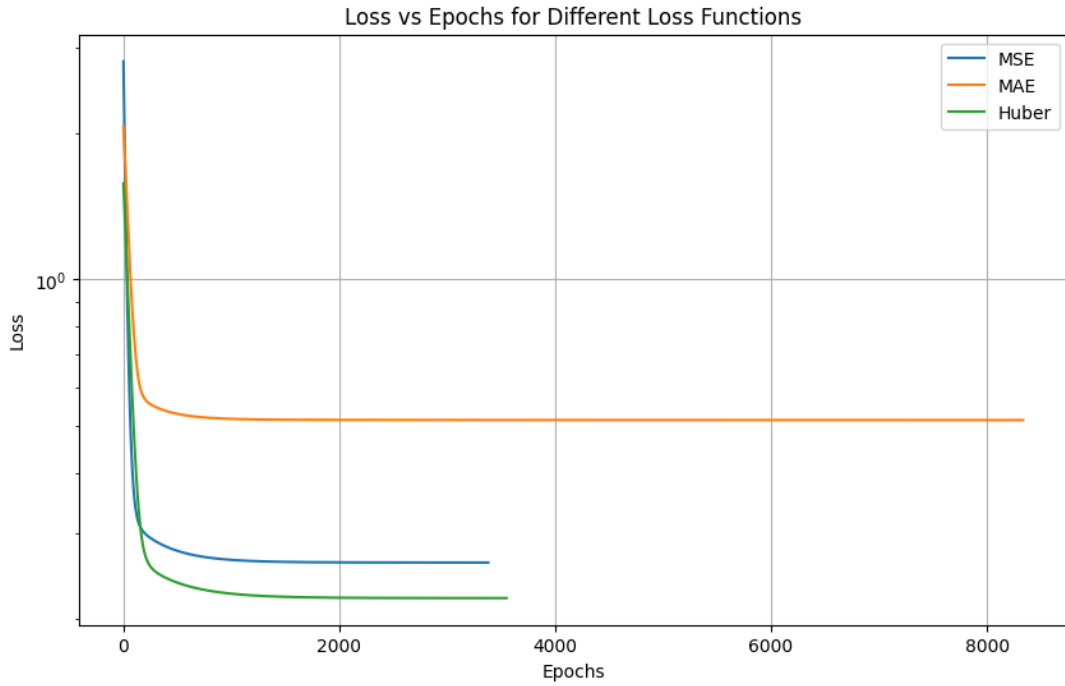


Figure 1: Loss vs Epochs for MSE, MAE, and Huber Loss

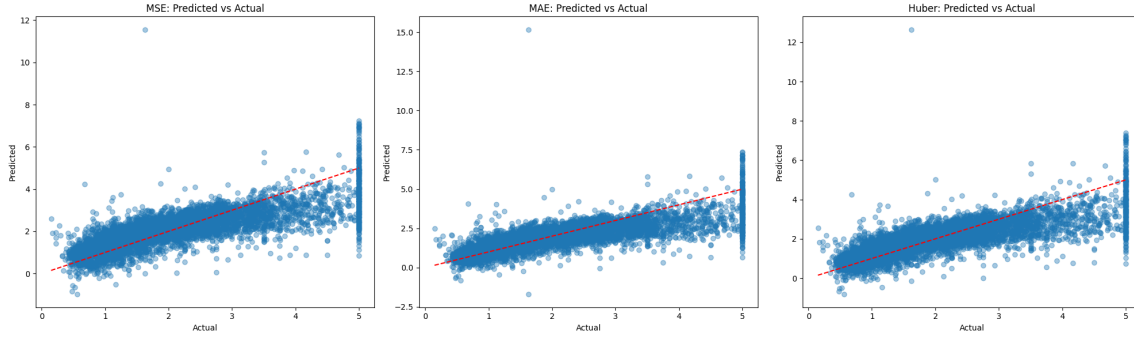


Figure 2: Predicted vs Actual Values for Different Loss Functions

5.1 Convergence Speed

Mean Squared Error (MSE) converges the fastest because it is smooth and continuously differentiable, providing gradients proportional to the prediction error. This allows large updates when errors are large and progressively smaller updates near the optimum, leading to stable and rapid convergence.

Mean Absolute Error (MAE) converges more slowly because its gradient depends only on the sign of the error and not its magnitude. As a result, parameter updates remain constant even near the minimum, causing oscillations around the optimum and slow convergence.

Huber loss shows intermediate behavior by acting like MSE for small errors and like MAE for large errors. This enables relatively fast convergence while maintaining stability, making it a practical compromise between speed and robustness.

5.2 Robustness to Outliers

Mean Squared Error (MSE) is highly sensitive to outliers because squaring the prediction error greatly amplifies large deviations. As a result, a small number of extreme data points can dominate the loss and gradient, causing the model to shift its parameters toward outliers and degrade overall performance.

Mean Absolute Error (MAE) is more robust to outliers because it penalizes errors linearly rather than quadratically. Large errors do not grow excessively, preventing outliers from having a disproportionate influence on training, although this robustness comes with slower optimization.

Huber loss provides a balance between robustness and efficiency by applying a quadratic penalty to small errors and a linear penalty to large errors. This limits the influence of outliers while preserving smooth gradients near the optimum, making Huber loss the most robust and practical choice in real-world regression tasks.

6 Conclusion

This study demonstrates that while Mean Squared Error offers the fastest convergence, it is highly sensitive to outliers. Mean Absolute Error provides robustness but suffers from slow convergence. Huber loss achieves the best trade-off between optimization efficiency and robustness, making it well-suited for real-world regression tasks.